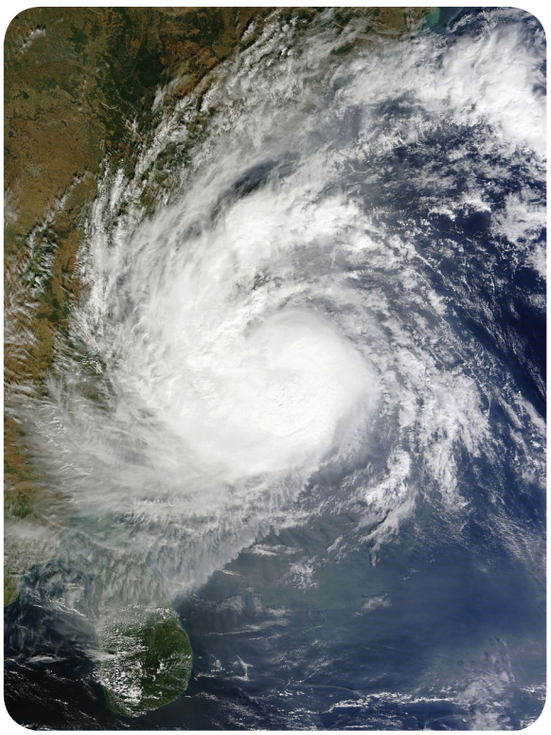


Figure 12.6 Very Severe Cyclonic Storm

meteorologist may be a good complement to other forecasting techniques.

At present, the cyclones are tracked with the help of satellite images. The satellite images are collected and put into motion for several hours, and with the gathered information, the development of a tropical cyclone is tracked. Doppler radars detect rain associated with cyclones, which locate and estimate the amount of rainfall and depicts a hurricane's rainfall.

In the satellite based techniques, track and intensity of cyclone are forecast based on the cloud pattern associated with the cyclone. Generally the outer cloud bands of cumulonimbus clouds indicate the future direction, and the cloud pattern surrounding eye of the cyclone indicates the future intensity of the cyclone. Below is an example of tracking Vardah Cyclone by using satellite images.

Cyclone Tracking based on Satellite Images – Vardah (Example)

Vardah at peak strength on 11 December

Formed	December 6, 2016
Dissipated	December 11, 2016
Highest Winds	3- minute sustained : 130 km/h 1-minute sustained : 155km/h
Lowest pressure	975 mb
Fatalities	38
Damage	$5.1 billion
Areas affected	Thailand, Sumatra, Malaysia, Andaman and Nicobar Islands, Sri Lanka, South India, Somalia

Very Severe Cyclonic Storm, Vardah, was the fourth cyclonic storm, as well as the most intense tropical cyclone of the 2016 North Indian Ocean. The system struck the Andaman and Nicobar Islands, as well as South India. Originating as a low pressure area near the Malay Peninsula on December 3, the storm was designated as a depression on December 6. It gradually intensified into a Deep Depression on the following day, skirting off the Andaman and Nicobar Islands, and intensified into a Cyclonic Storm on December 8. Maintaining a generally westward track thereafter, Vardah consolidated into a Severe Cyclonic Storm on December 9, before peaking as a Very Severe Cyclonic Storm, with winds of 130 km/hr, and a minimum central pressure of 982 mb on December 11. Weakening into a Severe Cyclonic Storm, Vardah, made landfall close to Chennai on the following day, and degenerated into a remnant low on December 13.The name *Vardah*, suggested by Pakistan, refers to the red rose.

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